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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create sample sales dataset
data = {
    'Date': ['2026-01-01', '2026-01-02', '2026-01-03',
            '2026-01-04', '2026-01-05', '2026-01-06',
            '2026-01-07', '2026-01-08', '2026-01-09'],
    'Product': ['Laptop', 'Mobile', 'Tablet',
               'Laptop', 'Mobile', 'Tablet',
               'Laptop', 'Mobile', 'Tablet'],
    'Sales': [50000, 20000, 15000, 55000, 25000,
              18000, 60000, 30000, 20000],
    'Quantity': [2, 4, 3, 2, 5, 4, 3, 6, 4],
    'Region': ['Chennai', 'Madurai', 'Trichy',
               'Chennai', 'Madurai', 'Trichy',
               'Chennai', 'Madurai', 'Trichy']
}

df = pd.DataFrame(data)

# Display the first five rows
print("First Five Rows:")
print(df.head())

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Handle missing values
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region', 'Date'],
          inplace=True)

# Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
df.dropna(subset=['Date'], inplace=True)

# Summary statistics
print("\nSummary Statistics:")
print(df.describe())

# Group by product
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print("\nProduct Summary:")
print(product_summary)

# Bar chart of total sales by product

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plt.figure(figsize=(8, 5))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.tight_layout()
plt.show()

# Line chart of sales over time
sales_over_time = df.groupby('Date')['Sales'].sum().reset_index()

plt.figure(figsize=(9, 5))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'],
         marker='o')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()

# Pivot table: sales by region and product
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print("\nSales by Region and Product:")
print(pivot_table)

# Correlation matrix
correlation_matrix = df.select_dtypes(
    include=np.number
).corr()

print("\nCorrelation Matrix:")
print(correlation_matrix)

# Heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(correlation_matrix, annot=True,
            cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()

print("\nSales Data Analysis Completed Successfully!")

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First Five Rows:

	Date	Product	Sales	Quantity	Region
0	2026-01-01	Laptop	50000	2	Chennai
1	2026-01-02	Mobile	20000	4	Madurai
2	2026-01-03	Tablet	15000	3	Trichy
3	2026-01-04	Laptop	55000	2	Chennai
4	2026-01-05	Mobile	25000	5	Madurai

Missing Values:

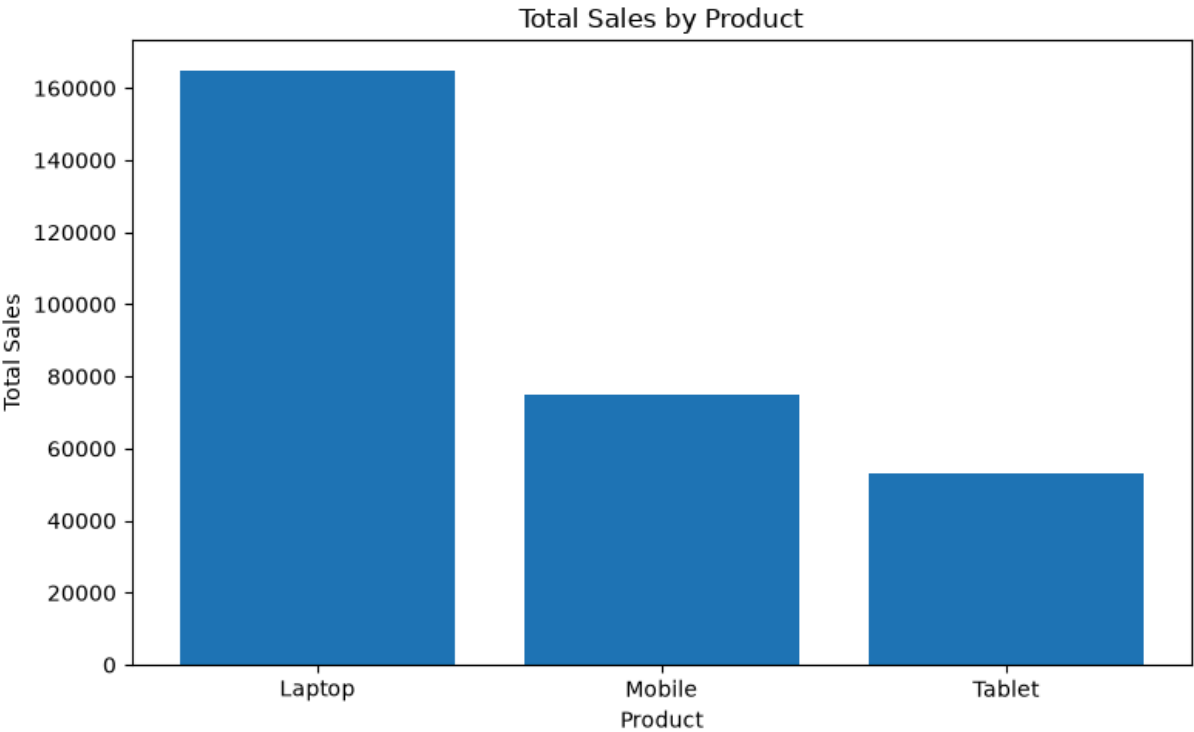
Date 0
Product 0
Sales 0
Quantity 0
Region 0
dtype: int64

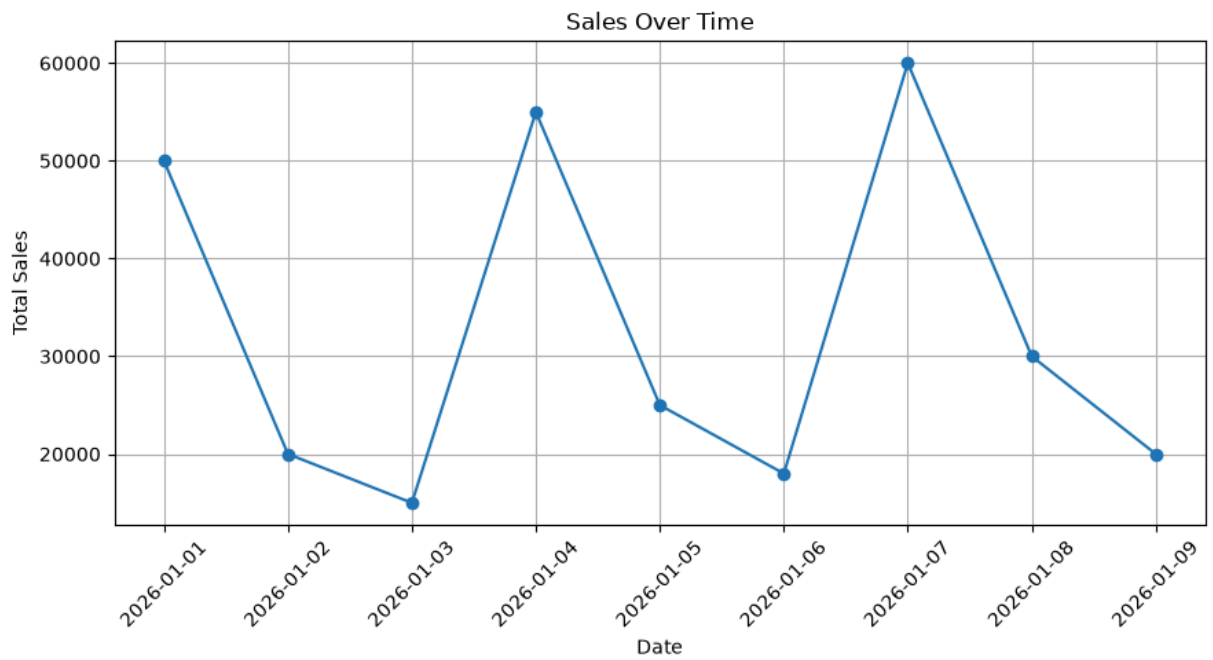
Summary Statistics:

	Date	Sales	Quantity
count	9	9.000000	9.000000
mean	2026-01-05 00:00:00	32555.555556	3.666667
min	2026-01-01 00:00:00	15000.000000	2.000000
25%	2026-01-03 00:00:00	20000.000000	3.000000
50%	2026-01-05 00:00:00	25000.000000	4.000000
75%	2026-01-07 00:00:00	50000.000000	4.000000
max	2026-01-09 00:00:00	60000.000000	6.000000
std	NaN	17536.469935	1.322876

Product Summary:

	Product	Sales	Quantity
0	Laptop	165000	7
1	Mobile	75000	15
2	Tablet	53000	11





Sales by Region and Product:

Product	Laptop	Mobile	Tablet
Region			
Chennai	165000	0	0
Madurai	0	75000	0
Trichy	0	0	53000

Correlation Matrix:

	Sales	Quantity
Sales	1.000000	-0.551399
Quantity	-0.551399	1.000000



Sales Data Analysis Completed Successfully!

In []: